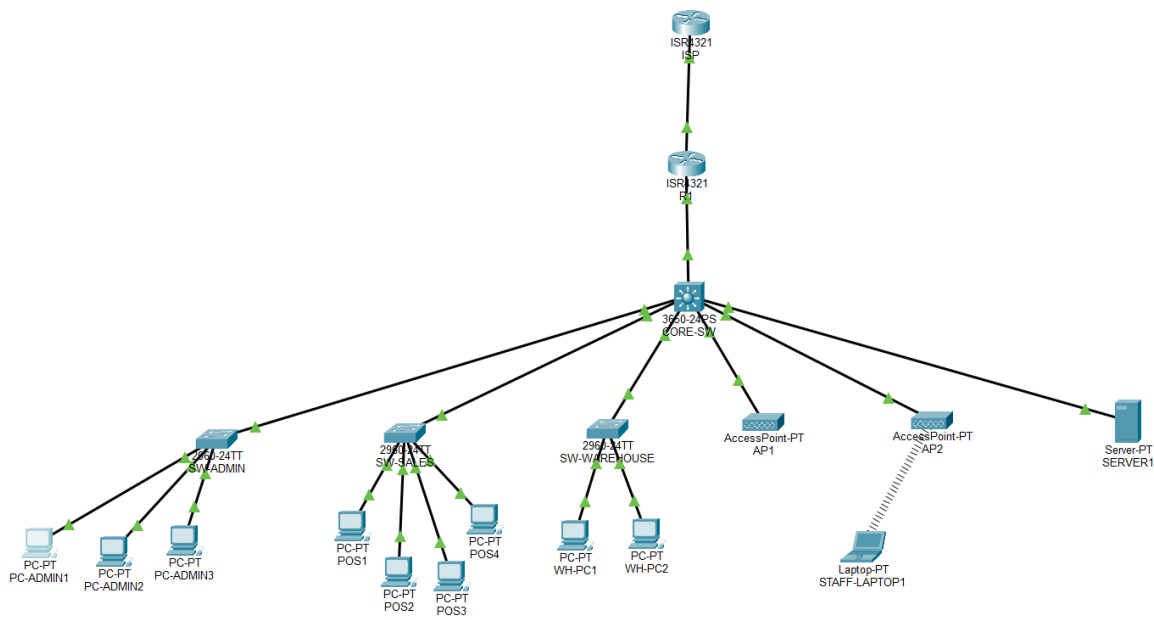


CMPG 325

Network Design Project

Tlou Furniture & Appliances (Rustenburg)

Phase 2 - Packet Tracer Implementation & Testing Report



Final implemented Cisco Packet Tracer topology and simulation view.

Client ID: CLI-017

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1. Project Overview

This report documents the Phase 2 Cisco Packet Tracer implementation for Tlou Furniture & Appliances (Rustenburg). The implementation was built from the Phase 1 network design using the assigned 172.30.2.0/23 address block. The completed network provides segmented connectivity for Admin, Sales/POS, Warehouse, wireless staff, network management and local server services, together with routing to a simulated ISP.

The Phase 1 brief required a working and testable Packet Tracer implementation, end-to-end connectivity, reuse of the core and access switches where possible and the assigned wireless-security feature using WPA2-PSK. The future branch block 172.30.3.0/24 remains reserved and was not built as a second site, in line with the change request.

1.1 Phase 2 objectives completed

- Built the physical and logical topology in Cisco Packet Tracer.
- Created departmental VLANs and a dedicated management VLAN.
- Configured inter-VLAN routing on the multilayer core switch.
- Configured 802.1Q trunks between the core and access switches.
- Configured DHCP for Admin, Sales/POS, Warehouse and Staff Wi-Fi clients.
- Configured a static server address in the server subnet.
- Implemented WPA2-PSK with AES on two staff wireless access points.
- Configured the edge router and ISP simulation for routed/NAT Internet testing.
- Verified local, inter-VLAN, server, wireless and simulated Internet connectivity

2. Implemented Topology

The final topology uses one ISR4321 edge router (R1), one ISR4321 simulated ISP router, one 3560-24PS multilayer core switch, three 2960 access switches, two AccessPoint-PT devices, one Server-PT, nine wired client devices and one wireless staff laptop. The 3560 performs Layer 3 switching and inter-VLAN routing.

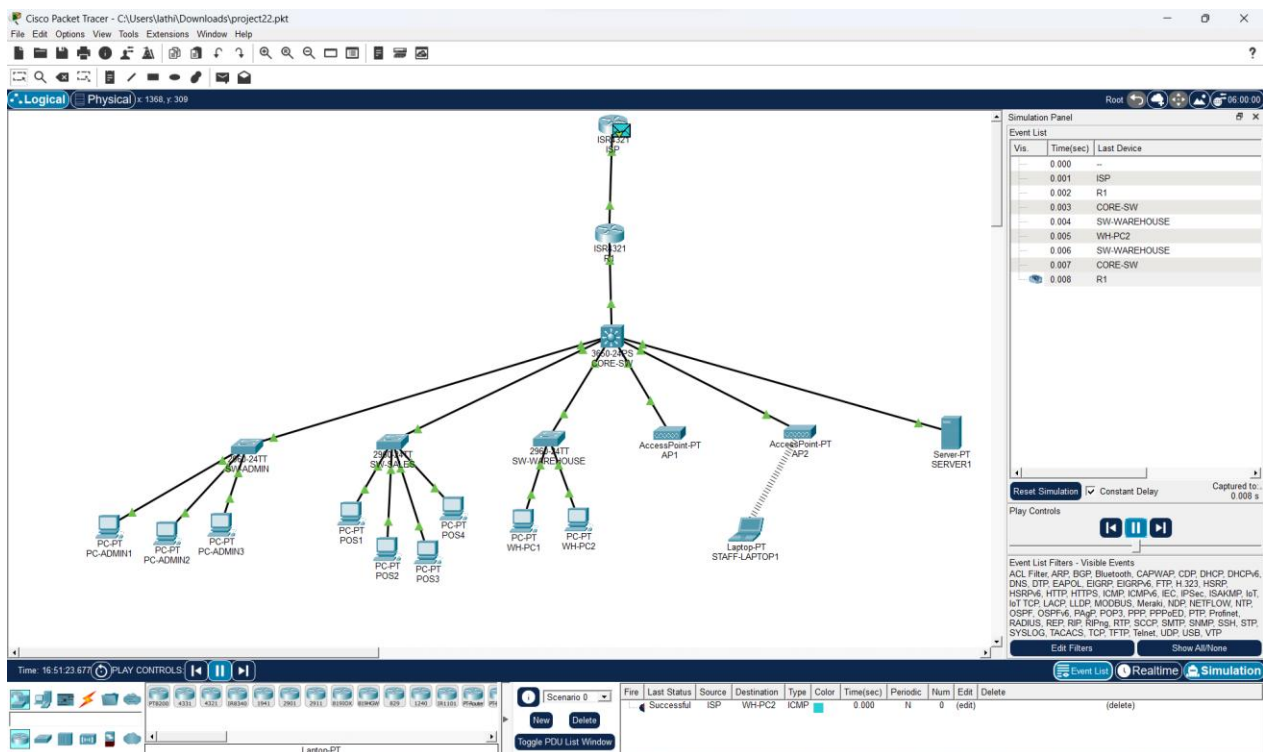


Figure 1. Final Packet Tracer topology. The simulation event list also shows successful packet traversal through ISP, R1, CORE-SW and the access layer.

Device	Role	Key connectivity
ISP	Simulated upstream network	Connected to R1; provides test destination 8.8.8.8
R1	Edge router / NAT	Inside: 172.30.2.189/30; outside: 172.30.2.185/30
CORE-SW	Layer 3 core/distribution	SVIs, DHCP, inter-VLAN routing and routed uplink to R1
SW-ADMIN	Admin access switch	VLAN 10 access ports; VLAN 99 management
SW-SALES	Sales/POS access switch	VLAN 20 access ports; VLAN 99 management
SW-WAREHOUSE	Warehouse access switch	VLAN 30 access ports; VLAN 99 management
AP1 / AP2	Staff wireless	VLAN 40, WPA2-PSK/AES, SSID Tlou-Staff-WiFi
SERVER1	Local POS/inventory server	172.30.2.178/29, gateway 172.30.2.177

3. VLAN and IP Addressing Plan

The addressing plan follows the Phase 1 VLSM design. VLAN 50 was used as the server VLAN identifier in the implementation because the Phase 1 document defines the server subnet but does not assign a VLAN number.

VLAN / Segment	Network	Mask	Gateway	Purpose
20 - SALES_POS	172.30.2.0/26	255.255.255.192	172.30.2.1	Sales and POS terminals
10 - ADMIN	172.30.2.64/27	255.255.255.224	172.30.2.65	Administration
30 - WAREHOUSE	172.30.2.96/27	255.255.255.224	172.30.2.97	Warehouse / inventory
40 - STAFF_WIFI	172.30.2.128/27	255.255.255.224	172.30.2.129	Staff wireless
99 - NETWORK_MANAGEMENT	172.30.2.160/28	255.255.255.240	172.30.2.161	Switch/AP management
50 - SERVERS	172.30.2.176/29	255.255.255.248	172.30.2.177	Local server segment
R1 - CORE transit	172.30.2.188/30	255.255.255.252	N/A	R1 .189 / CORE-SW .190
R1 - ISP transit	172.30.2.184/30	255.255.255.252	N/A	R1 .185 / ISP .186
Future branch reserve	172.30.3.0/24	255.255.255.0	Reserved	CR6 future branch

4. Configuration and Implementation

4.1 Device naming

Devices were renamed to make the topology readable and to match their business functions: R1, CORE-SW, SW-ADMIN, SW-SALES, SW-WAREHOUSE, AP1, AP2, SERVER1, PC-ADMIN1-3, POS1-4, WH-PC1-2 and STAFF-LAPTOP1.

4.2 Core switch VLANs and Layer 3 interfaces

VLANs 10, 20, 30, 40, 50 and 99 were created on CORE-SW. IP routing was enabled and an SVI was configured for each VLAN so the core switch could act as the default gateway and route between departmental networks.

```
ip routing
vlan 10
  name ADMIN
vlan 20
  name SALES_POS
vlan 30
  name WAREHOUSE
vlan 40
  name STAFF_WIFI
vlan 50
  name SERVERS
vlan 99
  name NETWORK_MANAGEMENT

interface vlan 10
  ip address 172.30.2.65 255.255.255.224
interface vlan 20
  ip address 172.30.2.1 255.255.255.192
interface vlan 30
  ip address 172.30.2.97 255.255.255.224
interface vlan 40
  ip address 172.30.2.129 255.255.255.224
interface vlan 50
  ip address 172.30.2.177 255.255.255.248
interface vlan 99
  ip address 172.30.2.161 255.255.255.240
```

Remark: Please zoom in to see images better.

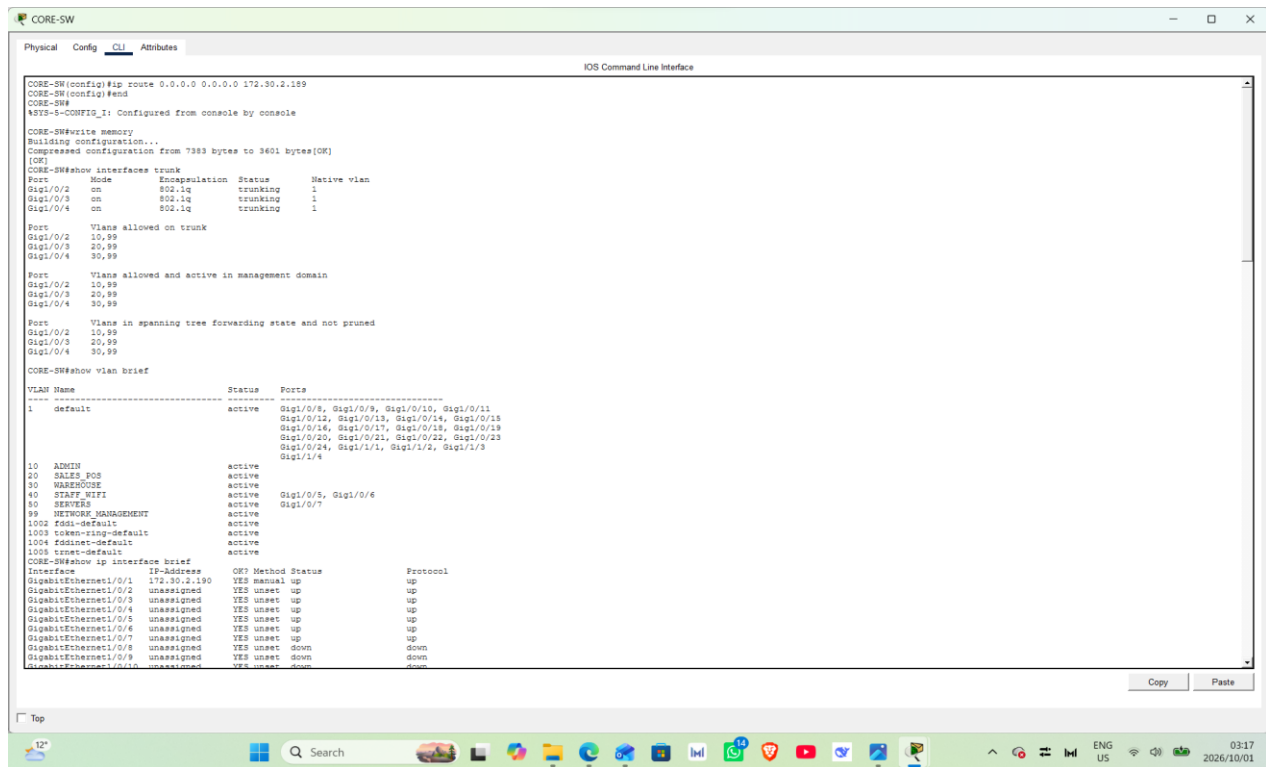


Figure 2. CORE-SW verification showing trunk ports, configured VLANs and the routed uplink address 172.30.2.190.

4.3 Trunk configuration and native VLAN correction

The core-to-access-switch links were configured as 802.1Q trunks. VLAN 99 was selected as the native and management VLAN. During configuration, CDP and STP reported native VLAN mismatch messages because the core had already been changed to native VLAN 99 while the access switches were still using the default native VLAN 1. The mismatch was corrected by setting VLAN 99 as the native VLAN on both ends of each trunk.

```

interface GigabitEthernet1/0/2
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,99

```

```

interface GigabitEthernet1/0/3
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 20,99

```

```

interface GigabitEthernet1/0/4
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 30,99

```

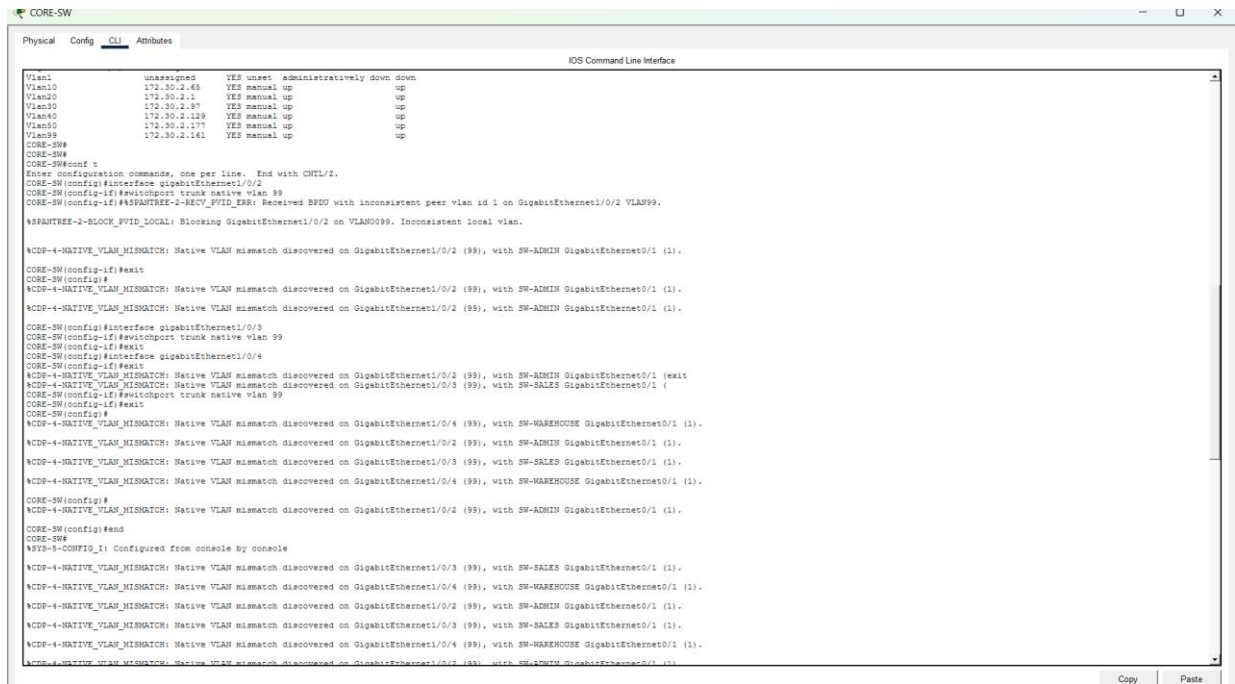


Figure 3. Native-VLAN mismatch messages observed during configuration before matching the access-switch trunk settings.

4.4 Access-switch configuration

Each 2960 access switch was configured with its departmental VLAN, VLAN 99 for management, access ports for end devices, a trunk uplink to CORE-SW and a management SVI in VLAN 99. PortFast was enabled on end-device access ports.

Switch	Access VLAN	End-device ports	Management IP	Trunk allows
SW-ADMIN	10	Fa0/1-3	172.30.2.162/28	10,99
SW-SALES	20	Fa0/1-4	172.30.2.163/28	20,99
SW-WAREHOUSE	30	Fa0/1-2	172.30.2.164/28	30,99

```
SW-ADMIN
Physical Config CLI Attributes
IOS Command Line Interface

SW-ADMIN(config-if)#switchport trunk allowed vlan 10,99
SW-ADMIN(config-if)#no shutdown
SW-ADMIN(config-if)#exit
SW-ADMIN(config)#interface vlan 99
SW-ADMIN(config-if)#
VLAN99-1-CHANGED: Interface Vlan99, changed state to up
VLAN99-1-UPDOWN: Line protocol on Interface Vlan99, changed state to up
SW-ADMIN(config-if)#ip address 172.30.2.162 255.255.255.240
SW-ADMIN(config-if)#no shutdown
SW-ADMIN(config-if)#exit
SW-ADMIN(config)#ip default-gateway 172.30.2.161
SW-ADMIN(config)#end
SW-ADMIN
VLAN99-1-CONFIG_1: Configured from console by console
SW-ADMIN#write memory
Building configuration...
[OK]
SW-ADMIN#show vlan br

VLAN Name      Status      Ports
-----
1  default      active      Fa0/4, Fa0/5, Fa0/6, Fa0/7,
                        Fa0/8, Fa0/9, Fa0/10, Fa0/11,
                        Fa0/12, Fa0/13, Fa0/14, Fa0/15,
                        Fa0/16, Fa0/17, Fa0/18, Fa0/19,
                        Fa0/20, Fa0/21, Fa0/22, Fa0/23,
                        Fa0/24, Gig0/2
10  ADMIN        active      Fa0/1, Fa0/2, Fa0/3
99  NETWORK_MANAGEMENT active
1002 fddi-default active
1003 token-ring-default active
1004 fdisknet-default active
1005 trnet-default active

SW-ADMIN#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    99

Port      Vlans allowed on trunk
Gig0/1    10,99

Port      Vlans allowed and active in management domain
Gig0/1    10,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,99

SW-ADMIN#show ip interface brief
Interface  IP-Address  OK? Method Status  Protocol
FastEthernet0/1  unassigned YES manual up
FastEthernet0/2  unassigned YES manual up
FastEthernet0/3  unassigned YES manual up
FastEthernet0/4  unassigned YES manual down
FastEthernet0/5  unassigned YES manual down
FastEthernet0/6  unassigned YES manual down
FastEthernet0/7  unassigned YES manual down
FastEthernet0/8  unassigned YES manual down
FastEthernet0/9  unassigned YES manual down
```

Figure 4. SW-ADMIN verification: VLAN 10 access ports, VLAN 99, 802.1Q trunk and management interface.

```
SW-SALES
Physical Config CLI Attributes
IOS Command Line Interface

SW-SALES(config-if)#no shutdown
SW-SALES(config-if)#exit
SW-SALES(config)#ip default-gateway 172.30.2.161
SW-SALES(config)#end
SW-SALES
VLAN99-1-CONFIG_1: Configured from console by console
SW-SALES#write memory
Building configuration...
[OK]
SW-SALES#show vlan brief

VLAN Name      Status      Ports
-----
1  default      active      Fa0/5, Fa0/6, Fa0/7, Fa0/8,
                        Fa0/9, Fa0/10, Fa0/11, Fa0/12,
                        Fa0/13, Fa0/14, Fa0/15, Fa0/16,
                        Fa0/17, Fa0/18, Fa0/19, Fa0/20,
                        Fa0/21, Fa0/22, Fa0/23, Fa0/24,
                        Gig0/2
20  SALES_POS    active      Fa0/1, Fa0/2, Fa0/3, Fa0/4
99  NETWORK_MANAGEMENT active
1002 fddi-default active
1003 token-ring-default active
1004 fdisknet-default active
1005 trnet-default active

SW-SALES#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    99

Port      Vlans allowed on trunk
Gig0/1    20,99

Port      Vlans allowed and active in management domain
Gig0/1    20,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    20,99

SW-SALES#show ip interface brief
Interface  IP-Address  OK? Method Status  Protocol
FastEthernet0/1  unassigned YES manual up
FastEthernet0/2  unassigned YES manual up
FastEthernet0/3  unassigned YES manual up
FastEthernet0/4  unassigned YES manual up
FastEthernet0/5  unassigned YES manual down
FastEthernet0/6  unassigned YES manual down
FastEthernet0/7  unassigned YES manual down
FastEthernet0/8  unassigned YES manual down
FastEthernet0/9  unassigned YES manual down
FastEthernet0/10 unassigned YES manual down
FastEthernet0/11 unassigned YES manual down
FastEthernet0/12 unassigned YES manual down
FastEthernet0/13 unassigned YES manual down
FastEthernet0/14 unassigned YES manual down
FastEthernet0/15 unassigned YES manual down
FastEthernet0/16 unassigned YES manual down
FastEthernet0/17 unassigned YES manual down
FastEthernet0/18 unassigned YES manual down
FastEthernet0/19 unassigned YES manual down
FastEthernet0/20 unassigned YES manual down
FastEthernet0/21 unassigned YES manual down
```

Figure 5. SW-SALES verification: VLAN 20 on Fa0/1-4 and trunk carrying VLANs 20 and 99.

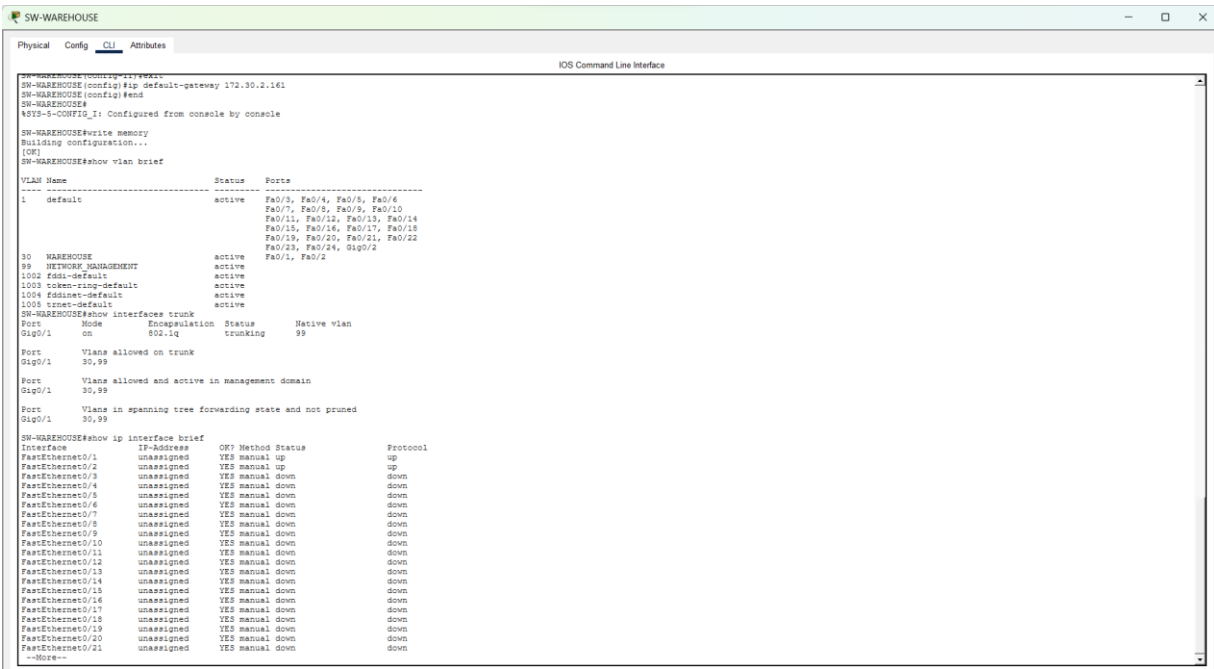


Figure 6. SW-WAREHOUSE verification: VLAN 30 on Fa0/1-2 and trunk carrying VLANs 30 and 99.

4.5 DHCP services

CORE-SW was configured as the DHCP server for wired clients and staff wireless clients. Gateway addresses and a small block of addresses at the start of each subnet were excluded from dynamic allocation. DNS was set to 8.8.8.8 for testing.

```

ip dhcp excluded-address 172.30.2.1 172.30.2.10
ip dhcp excluded-address 172.30.2.65 172.30.2.70
ip dhcp excluded-address 172.30.2.97 172.30.2.102
ip dhcp excluded-address 172.30.2.129 172.30.2.134

```

```

ip dhcp pool SALES
 network 172.30.2.0 255.255.255.192
 default-router 172.30.2.1
 dns-server 8.8.8.8
ip dhcp pool ADMIN
 network 172.30.2.64 255.255.255.224
 default-router 172.30.2.65
 dns-server 8.8.8.8
ip dhcp pool WAREHOUSE
 network 172.30.2.96 255.255.255.224
 default-router 172.30.2.97
 dns-server 8.8.8.8
ip dhcp pool STAFF-WIFI
 network 172.30.2.128 255.255.255.224
 default-router 172.30.2.129
 dns-server 8.8.8.8

```

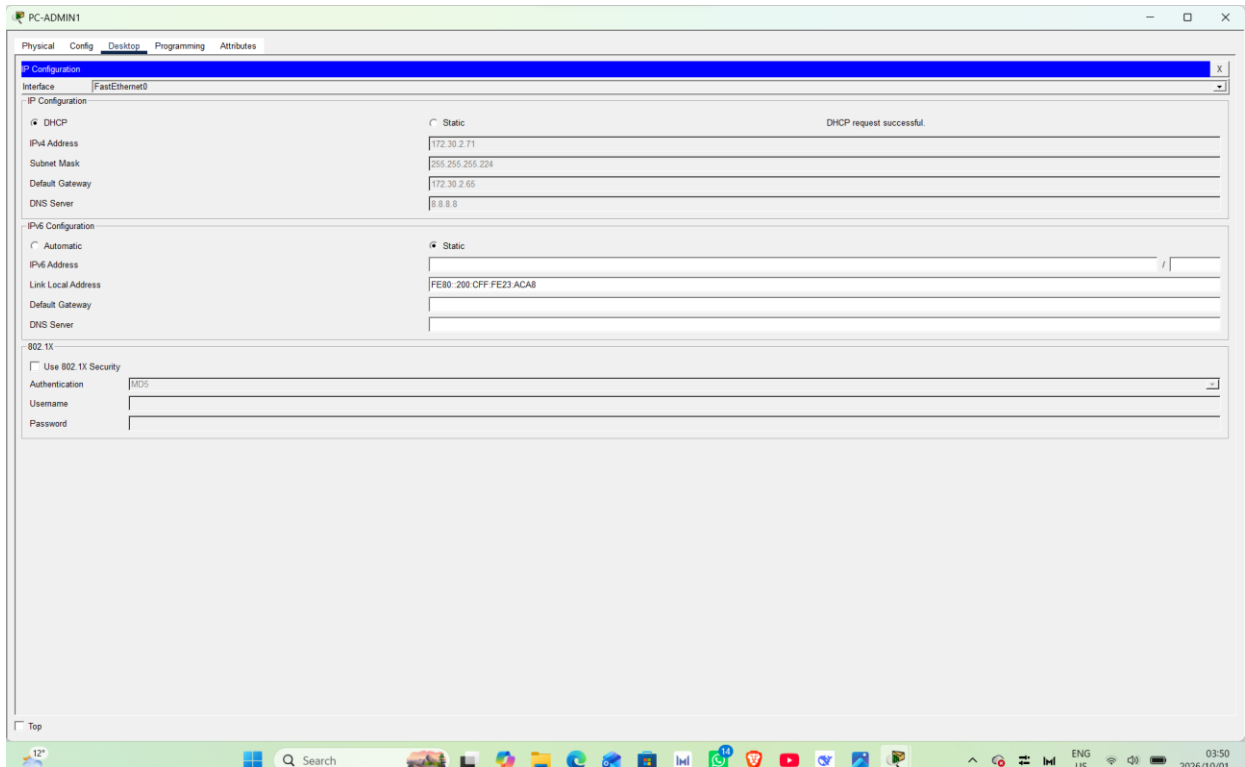


Figure 7. Admin client received 172.30.2.71/27, gateway 172.30.2.65 and DNS 8.8.8.8 via DHCP.

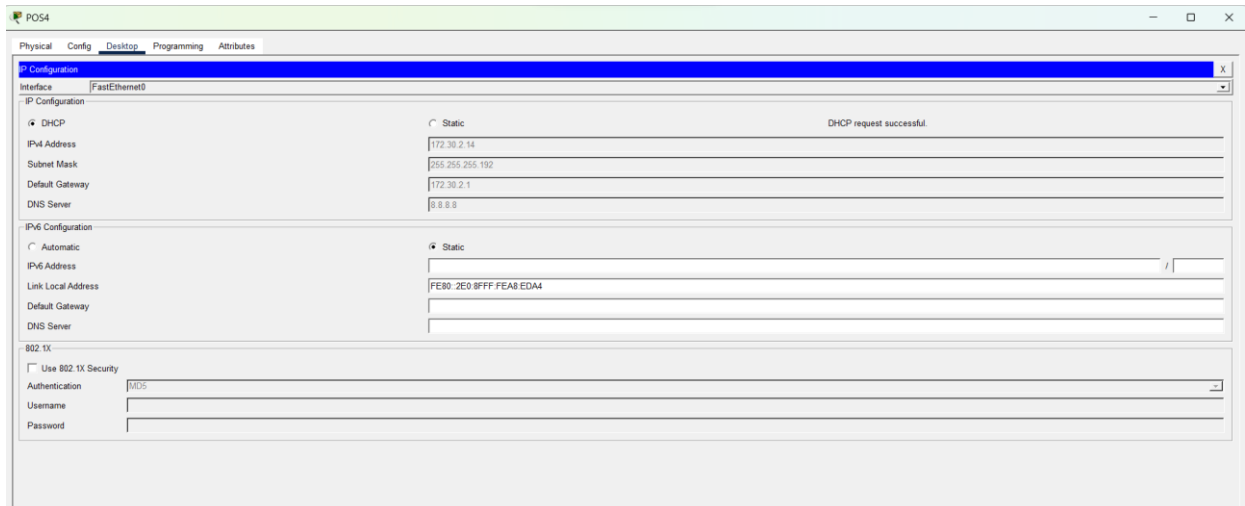


Figure 8. POS4 received 172.30.2.14/26, gateway 172.30.2.1 via DHCP.

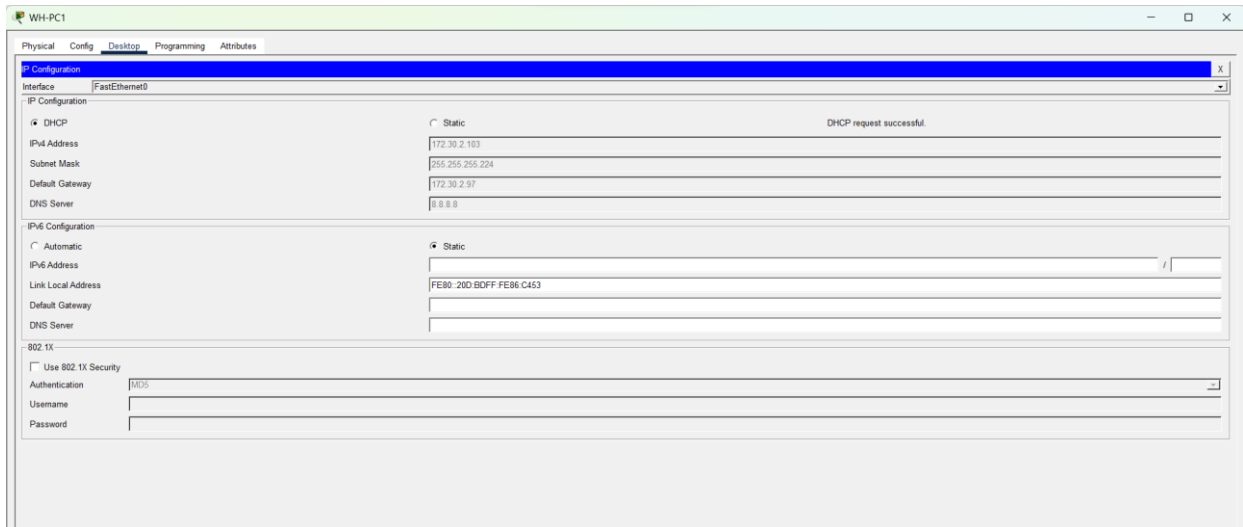


Figure 9. WH-PC1 received 172.30.2.103/27, gateway 172.30.2.97 via DHCP.

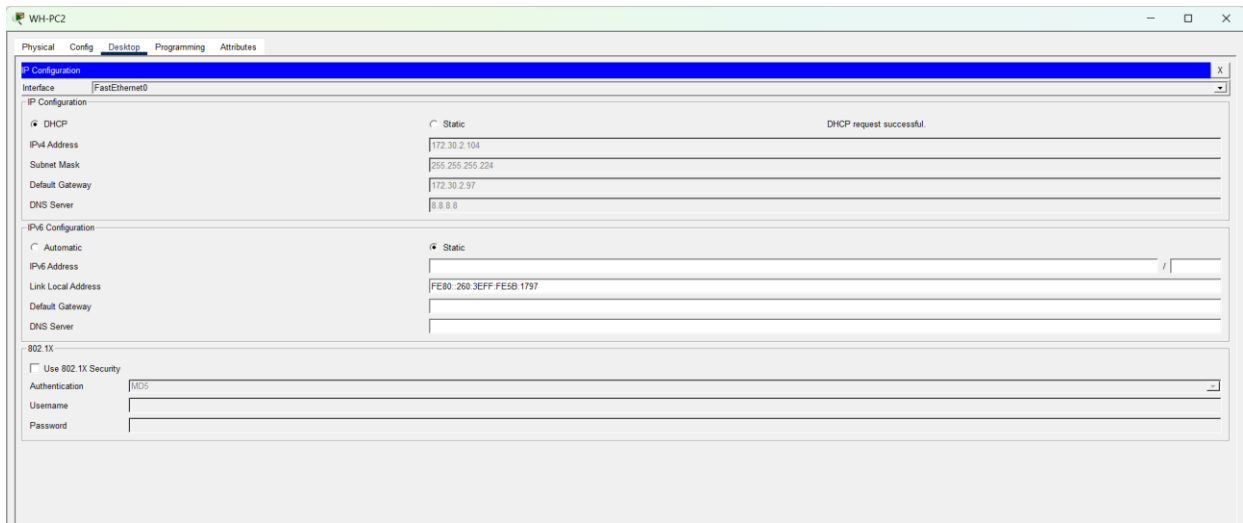


Figure 10. WH-PC2 received 172.30.2.104/27, gateway 172.30.2.97 via DHCP.

4.6 Server configuration

SERVER1 was configured with a static address because the local POS/inventory server must remain reachable at a predictable IP address. It was placed on the server VLAN using 172.30.2.178/29 with default gateway 172.30.2.177.

The screenshot shows the configuration window for SERVER1. The 'Configuration' tab is active, displaying the 'IP Configuration' section. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with the following values: IPv4 Address: 172.30.2.178, Subnet Mask: 255.255.255.248, Default Gateway: 172.30.2.177, and DNS Server: 8.8.8.8. The 'IPv6 Configuration' section is also visible, with 'Static' selected and a Link Local Address of FE80::203:E4FF:FED7:D862. The '802.1X' section is expanded, showing 'Use 802.1X Security' as unchecked, with fields for Authentication (RADIUS), Username, and Password.

Figure 11. SERVER1 static IPv4 configuration.

4.7 Wireless security implementation

The assigned networking feature was implemented on AP1 and AP2. Both access points use the same SSID so staff devices can use the staff WLAN. WPA2-PSK authentication and AES encryption were selected and the access-point switch ports were placed in VLAN 40.

Setting	Configured value
SSID	Tlou-Staff-WiFi
Security	WPA2-PSK
Encryption	AES
PSK	TlouStaff@2026
Wireless VLAN	40
VLAN 40 gateway	172.30.2.129

The screenshot shows the configuration window for AP1. The 'Config' tab is active, displaying the 'Port 1' configuration. The 'Port Status' is 'On'. The 'SSID' is 'Tlou-Staff-WiFi'. The '2.4 GHz Channel' is '6' and the 'Coverage Range (meters)' is '140.00'. Under 'Authentication', 'WPA2-PSK' is selected. The 'Encryption Type' is 'AES'. The 'WEP Key' field is empty. The 'PSK Pass Phrase' is 'TlouStaff@2026'. The 'User ID' and 'Password' fields are empty.

Figure 12. AP1 configured for SSID Tlou-Staff-WiFi with WPA2-PSK and AES.

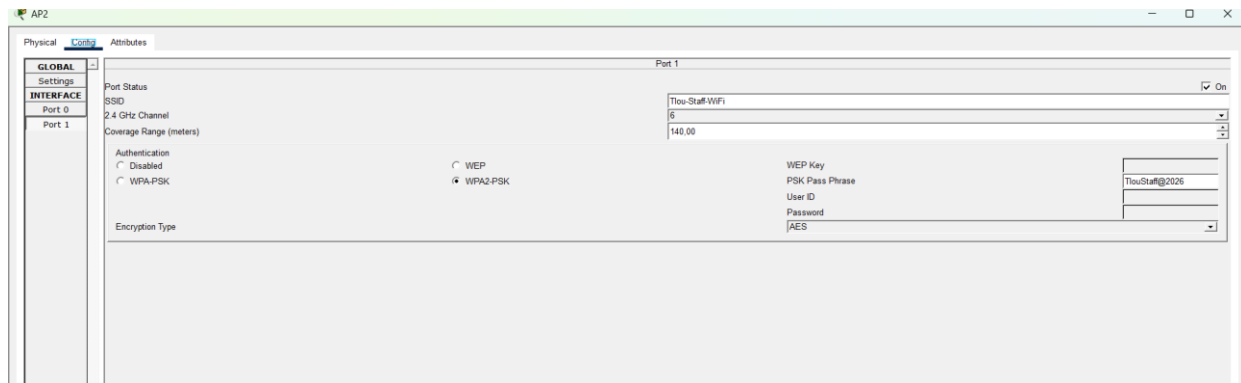


Figure 13. AP2 configured with the same WPA2-PSK/AES staff WLAN settings.

4.8 R1, ISP and NAT configuration

The edge router provides the routed link between CORE-SW and the simulated ISP. During troubleshooting, CDP showed that the physical R1 interfaces were the reverse of the initial assumption. The addresses were therefore placed on the interfaces actually connected to each neighbor: Gi0/0/1 toward CORE-SW and Gi0/0/0 toward ISP. NAT overload was applied on the ISP-facing interface.

```
interface GigabitEthernet0/0/1
description TO-CORE-SW
ip address 172.30.2.189 255.255.255.252
ip nat inside
no shutdown
```

```
interface GigabitEthernet0/0/0
description TO-ISP
ip address 172.30.2.185 255.255.255.252
ip nat outside
no shutdown
```

```
ip route 172.30.2.0 255.255.254.0 172.30.2.190
ip route 0.0.0.0 0.0.0.0 172.30.2.186
access-list 1 permit 172.30.2.0 0.0.1.255
ip nat inside source list 1 interface GigabitEthernet0/0/0 overload
```

```
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0 172.30.2.185    YES manual up          up
GigabitEthernet0/0/1 172.30.2.189    YES manual up          up
Vlan1          unassigned      YES unset  up          down
R1#
```

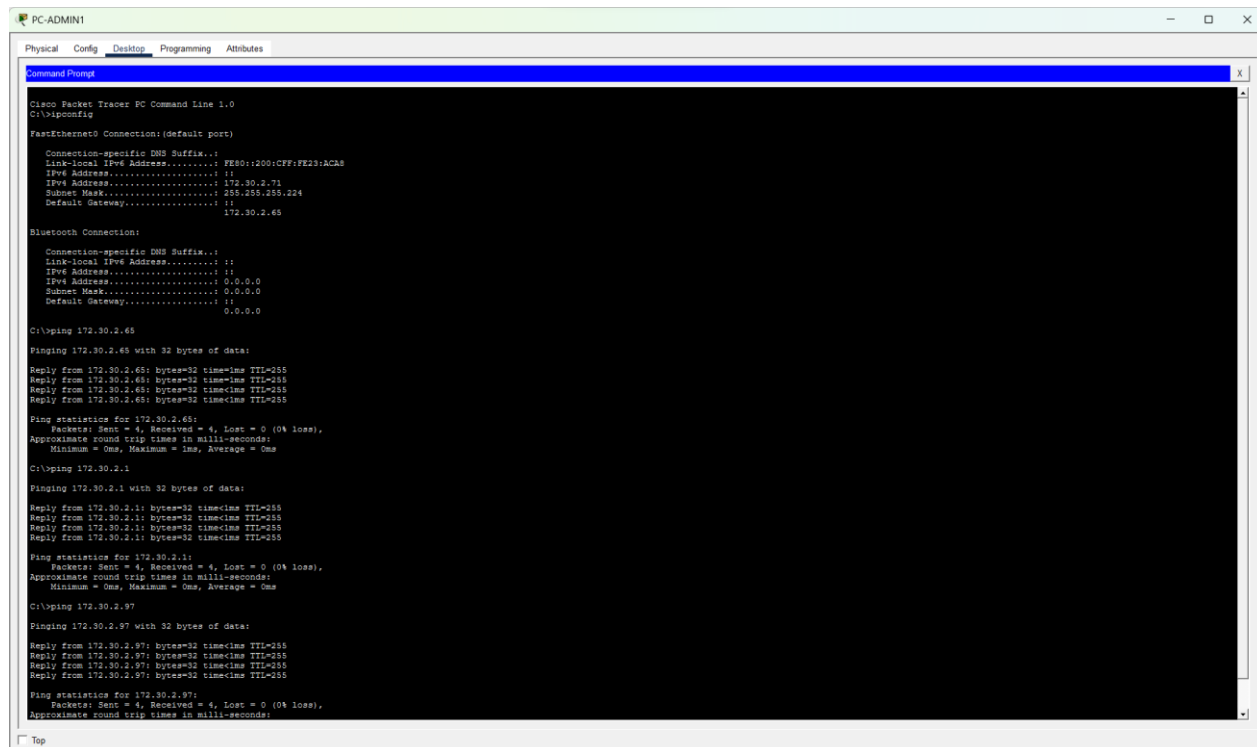
Figure 14. R1 interface verification after correcting the physical interface mapping: Gi0/0/0 = 172.30.2.185 and Gi0/0/1 = 172.30.2.189, both up/up.

5. Testing and Verification Evidence

Testing was carried out at several layers: VLAN/trunk verification, DHCP address allocation, default-gateway reachability, inter-VLAN routing, server reachability, wireless connectivity, edge routing, and simulated Internet reachability. The screenshots below provide direct Packet Tracer evidence.

5.1 Inter-VLAN routing from an Admin client

PC-ADMIN1 received 172.30.2.71/27 and successfully reached the Admin gateway (172.30.2.65), Sales gateway (172.30.2.1) and Warehouse gateway (172.30.2.97), demonstrating DHCP and routing between VLANs.



```
PC-ADMIN1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::200:CFF:FE23:ACAB
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.30.2.71
    Subnet Mask . . . . .: 255.255.255.224
    Default Gateway . . . . .: 172.30.2.65

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .: 0.0.0.0
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 172.30.2.65

Pinging 172.30.2.65 with 32 bytes of data:

Reply from 172.30.2.65: bytes=32 time=1ms TTL=255
Reply from 172.30.2.65: bytes=32 time=1ms TTL=255
Reply from 172.30.2.65: bytes=32 time=1ms TTL=255
Reply from 172.30.2.65: bytes=32 time=1ms TTL=255

Ping statistics for 172.30.2.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.30.2.1

Pinging 172.30.2.1 with 32 bytes of data:

Reply from 172.30.2.1: bytes=32 time=1ms TTL=255
Reply from 172.30.2.1: bytes=32 time=1ms TTL=255
Reply from 172.30.2.1: bytes=32 time=1ms TTL=255
Reply from 172.30.2.1: bytes=32 time=1ms TTL=255

Ping statistics for 172.30.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.2.97

Pinging 172.30.2.97 with 32 bytes of data:

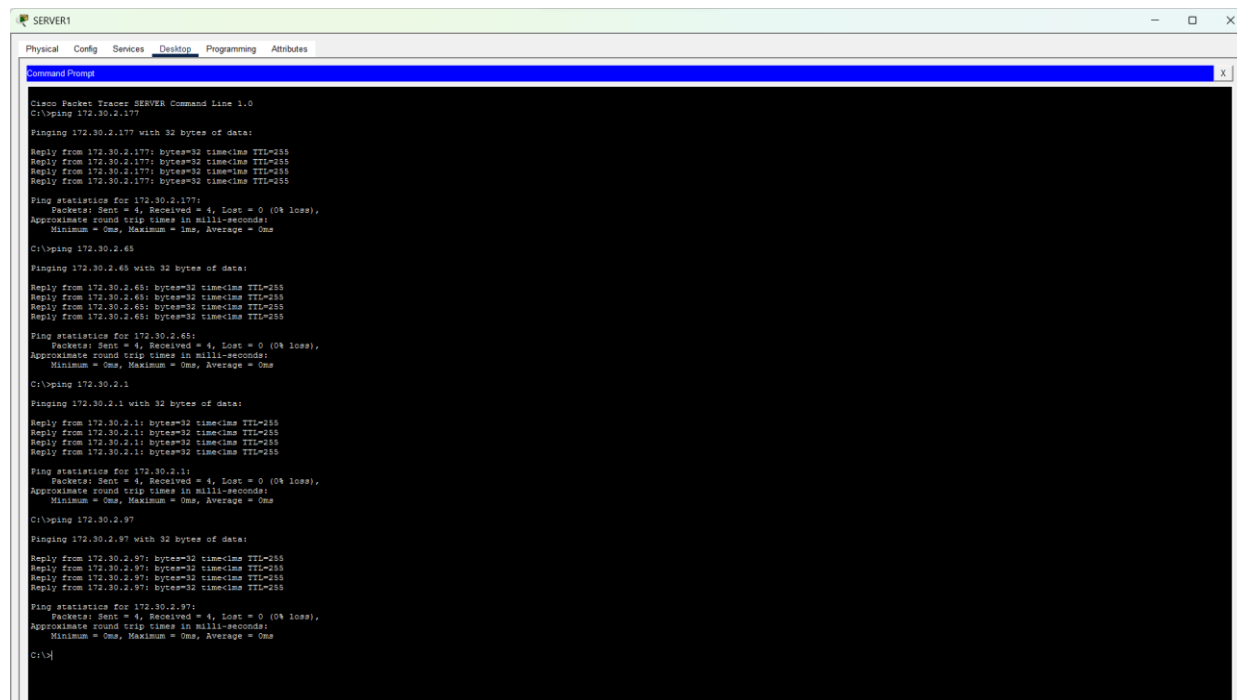
Reply from 172.30.2.97: bytes=32 time=1ms TTL=255
Reply from 172.30.2.97: bytes=32 time=1ms TTL=255
Reply from 172.30.2.97: bytes=32 time=1ms TTL=255
Reply from 172.30.2.97: bytes=32 time=1ms TTL=255

Ping statistics for 172.30.2.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

Figure 15. PC-ADMIN1 ipconfig and successful gateway/inter-VLAN ping tests with 0% packet loss.

5.2 Server connectivity

SERVER1 successfully reached its own VLAN gateway and the Admin, Sales and Warehouse gateways. This verifies that the server VLAN is routed through CORE-SW and can communicate with the departmental networks.



```
SERVER1
Physical Config Services Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 172.30.2.177

Pinging 172.30.2.177 with 32 bytes of data:

Reply from 172.30.2.177: bytes=32 time=1ms TTL=255
Reply from 172.30.2.177: bytes=32 time=1ms TTL=255
Reply from 172.30.2.177: bytes=32 time=1ms TTL=255
Reply from 172.30.2.177: bytes=32 time=1ms TTL=255

Ping statistics for 172.30.2.177:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.30.2.65

Pinging 172.30.2.65 with 32 bytes of data:

Reply from 172.30.2.65: bytes=32 time=1ms TTL=255
Reply from 172.30.2.65: bytes=32 time=1ms TTL=255
Reply from 172.30.2.65: bytes=32 time=1ms TTL=255
Reply from 172.30.2.65: bytes=32 time=1ms TTL=255

Ping statistics for 172.30.2.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.2.1

Pinging 172.30.2.1 with 32 bytes of data:

Reply from 172.30.2.1: bytes=32 time=1ms TTL=255
Reply from 172.30.2.1: bytes=32 time=1ms TTL=255
Reply from 172.30.2.1: bytes=32 time=1ms TTL=255
Reply from 172.30.2.1: bytes=32 time=1ms TTL=255

Ping statistics for 172.30.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.2.97

Pinging 172.30.2.97 with 32 bytes of data:

Reply from 172.30.2.97: bytes=32 time=1ms TTL=255
Reply from 172.30.2.97: bytes=32 time=1ms TTL=255
Reply from 172.30.2.97: bytes=32 time=1ms TTL=255
Reply from 172.30.2.97: bytes=32 time=1ms TTL=255

Ping statistics for 172.30.2.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 16. SERVER1 successful pings to 172.30.2.177, 172.30.2.65, 172.30.2.1 and 172.30.2.97.

5.3 Wireless connectivity

STAFF-LAPTOP1 connected through the staff wireless network and successfully pinged the VLAN 40 gateway (172.30.2.129), SERVER1 (172.30.2.178) and the Admin gateway (172.30.2.65). These tests demonstrate that the WPA2-secured wireless segment is operational and inter-VLAN routing is working for wireless clients.



```
STAFF-LAPTOP1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>

ping 172.30.2.129

Pinging 172.30.2.129 with 32 bytes of data:
Reply from 172.30.2.129: bytes=32 time=46ms TTL=255
Reply from 172.30.2.129: bytes=32 time=46ms TTL=255
Reply from 172.30.2.129: bytes=32 time=46ms TTL=255
Reply from 172.30.2.129: bytes=32 time=17ms TTL=255

Ping statistics for 172.30.2.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 66ms, Average = 31ms

C:\>ping 172.30.2.178

Pinging 172.30.2.178 with 32 bytes of data:
Reply from 172.30.2.178: bytes=32 time=13ms TTL=127
Reply from 172.30.2.178: bytes=32 time=22ms TTL=127
Reply from 172.30.2.178: bytes=32 time=25ms TTL=127
Reply from 172.30.2.178: bytes=32 time=23ms TTL=127

Ping statistics for 172.30.2.178:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 25ms, Average = 23ms

C:\>ping 172.30.2.65

Pinging 172.30.2.65 with 32 bytes of data:
Reply from 172.30.2.65: bytes=32 time=32ms TTL=255
Reply from 172.30.2.65: bytes=32 time=34ms TTL=255
Reply from 172.30.2.65: bytes=32 time=13ms TTL=255
Reply from 172.30.2.65: bytes=32 time=30ms TTL=255

Ping statistics for 172.30.2.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 34ms, Average = 27ms

C:\>
```

Figure 17. STAFF-LAPTOP1 successful wireless connectivity tests with 0% packet loss.

5.4 Edge and simulated Internet connectivity

After correcting the R1 interface mapping, PC-ADMIN1 successfully pinged the ISP-side address 172.30.2.186 and the simulated Internet test address 8.8.8.8. Both tests returned 0% packet loss, confirming the complete path through the access layer, CORE-SW, R1 and ISP.

```
C:\>ping 172.30.2.186

Pinging 172.30.2.186 with 32 bytes of data:

Reply from 172.30.2.186: bytes=32 time<1ms TTL=253
Reply from 172.30.2.186: bytes=32 time=1ms TTL=253
Reply from 172.30.2.186: bytes=32 time<1ms TTL=253
Reply from 172.30.2.186: bytes=32 time<1ms TTL=253

Ping statistics for 172.30.2.186:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 18. Successful pings from PC-ADMIN1 to 172.30.2.186 and 8.8.8.8 with 0% packet loss.

5.5 Simulation-mode evidence

Packet Tracer simulation mode was also used to observe packet traversal. The event list shows packet processing through ISP, R1, CORE-SW, SW-WAREHOUSE and WH-PC2 and back through the routed path, providing an additional visual confirmation of forwarding behavior.

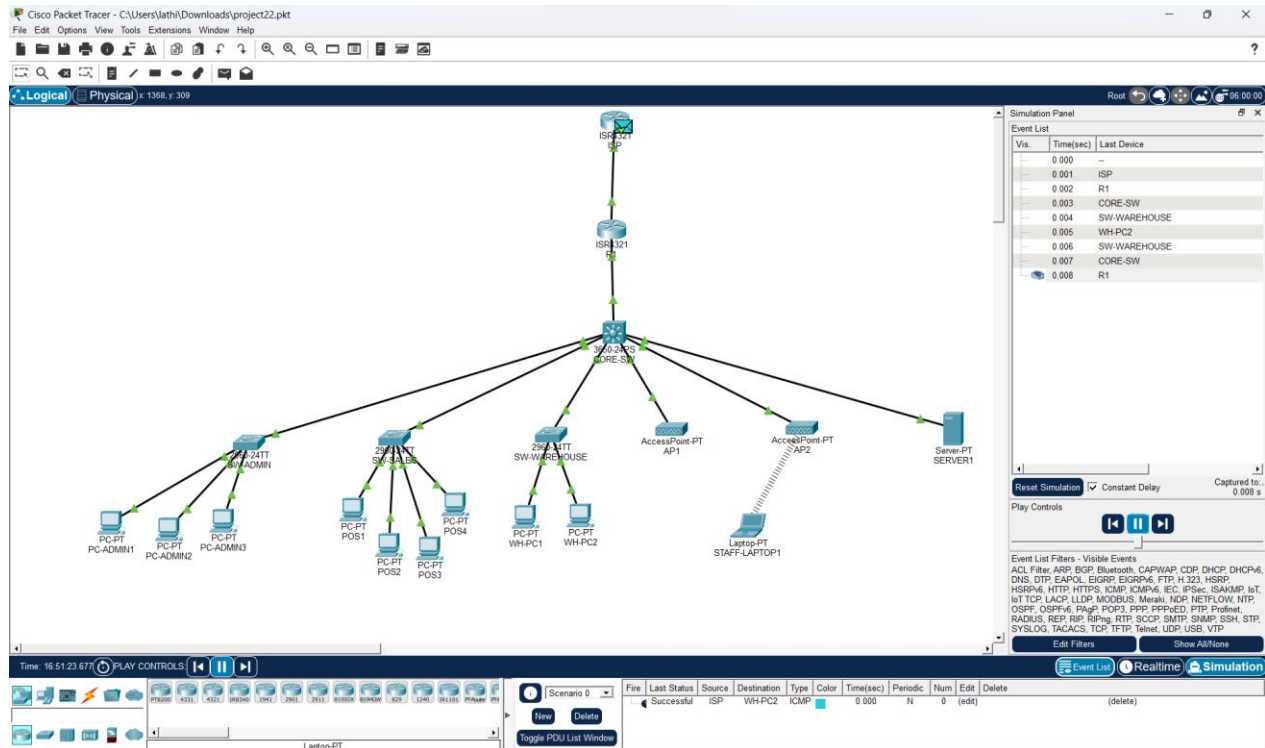


Figure 19. Packet Tracer simulation event list showing packet traversal across the completed topology.

6. Test Results Summary

Test	Expected result	Observed result	Status
VLAN creation	VLANs 10,20,30,40,50,99 active	VLANs present on CORE-SW	PASS
802.1Q trunks	Correct VLANs carried	Admin 10/99; Sales 20/99; Warehouse 30/99	PASS
DHCP - Admin	Client in 172.30.2.64/27	PC-ADMIN1 = 172.30.2.71	PASS
DHCP - Sales	Client in 172.30.2.0/26	POS4 = 172.30.2.14	PASS
DHCP - Warehouse	Clients in 172.30.2.96/27	WH-PC1=103; WH-PC2=104	PASS
Server static IP	172.30.2.178/29	Configured correctly	PASS
Inter-VLAN routing	Gateways reachable across VLANs	0% loss in shown tests	PASS
WPA2-PSK wireless	Secure staff SSID on AP1/AP2	WPA2-PSK + AES configured	PASS
Wireless routing	Wireless client reaches server/Admin	Successful pings	PASS
ISP reachability	172.30.2.186 reachable	0% loss	PASS
Simulated Internet	8.8.8.8 reachable	0% loss	PASS

7. Problems Encountered and Resolutions

Issue	Cause	Resolution
CORE-SW CLI initially unavailable	Multilayer switch was powered off	Powered on the device from the Physical tab before configuration.
Native VLAN mismatch / STP inconsistent port messages	CORE-SW trunks used native VLAN 99 while access-switch uplinks were still native VLAN 1	Configured native VLAN 99 on Gi0/1 of each access switch. Port consistency was restored.
STAFF-LAPTOP1 had no PC Wireless option	Laptop had only a FastEthernet module installed	Installed a compatible wireless NIC module in the Physical tab, rebooted, then connected to the staff SSID.
Cloud-PT had no CLI	Cloud-PT is not an IOS router	Replaced the cloud role with an ISR4321 simulated ISP router.
PC could reach LAN but not R1/ISP	R1 physical interfaces were assigned opposite to actual cable connections	Used CDP to identify neighbors, then moved .189 to Gi0/0/1 (CORE-SW) and .185 to Gi0/0/0 (ISP), with NAT inside/outside corrected.

8. Wireless Security Feature

The assigned feature, wireless security hardening, was implemented using WPA2-PSK and AES. The staff wireless network is separated from the wired departmental VLANs in VLAN 40. The configured SSID is Tlou-Staff-WiFi on both AP1 and AP2. The evidence in Figures 12 and 13 confirms that WPA2-PSK is selected and AES is used for encryption. This satisfies the Phase 1 requirement to demonstrate WPA2-PSK hardening in Packet Tracer.

9. Conclusion

The Phase 2 Packet Tracer implementation is operational and testable. Departmental VLANs, management segmentation, Layer 3 routing, DHCP, static server addressing, secured staff wireless access, edge routing, NAT and simulated Internet access were configured and verified. The attached evidence demonstrates successful address allocation and 0% packet-loss connectivity across the principal LAN, wireless, server and upstream paths. The future 172.30.3.0/24 branch allocation remains reserved for the CR6 design requirement and is not deployed as a second site in this phase.

Appendix A - Key Configuration Summary

Device / Function	Key setting
CORE-SW routed uplink	Gi1/0/1 = 172.30.2.190/30; default route via 172.30.2.189
CORE-SW AP ports	Gi1/0/5-6 access VLAN 40
CORE-SW server port	Gi1/0/7 access VLAN 50
SW-ADMIN	Fa0/1-3 VLAN 10; Gi0/1 trunk native 99; mgmt 172.30.2.162/28
SW-SALES	Fa0/1-4 VLAN 20; Gi0/1 trunk native 99; mgmt 172.30.2.163/28
SW-WAREHOUSE	Fa0/1-2 VLAN 30; Gi0/1 trunk native 99; mgmt 172.30.2.164/28
R1 inside	Gi0/0/1 = 172.30.2.189/30
R1 outside	Gi0/0/0 = 172.30.2.185/30
ISP	172.30.2.186/30; simulated 8.8.8.8 destination
SERVER1	172.30.2.178/29; gateway 172.30.2.177
Wireless	SSID Tlou-Staff-WiFi; WPA2-PSK; AES; VLAN 40

Appendix B - Additional Configuration Evidence

The following screenshots show intermediate access-switch configuration steps and the correction of the trunk/native-VLAN consistency issue. They are included for completeness as evidence of the implementation process.

```
SW-ADMIN
Physical Config CLI Attributes
IOS Command Line Interface

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#exit
% Incomplete command.
Switch(config)#hostname SW-ADMIN
SW-ADMIN(config)#end
SW-ADMIN#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
SW-ADMIN#

SW-ADMIN con0 is now available

Press RETURN to get started.

%LINE-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SPANTREE-2-RECV_EVID_ERR: Received 802.1Q BPDU on non trunk GigabitEthernet0/1 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SPANTREE-2-RECV_EVID_ERR: Received BPDU with inconsistent peer vlan id 99 on GigabitEthernet0/1 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent local vlan.

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
```

B1. SW-ADMIN naming and early native-VLAN/STP messages.

```
SW-ADMIN
Physical Config CLI Attributes
IOS Command Line Interface

SW-ADMIN>
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
SW-ADMIN#en
SW-ADMIN#conf t
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
Enter configuration commands, one per line. End with CNTL/Z.
SW-ADMIN(config)#vlan 10
SW-ADMIN(config-vlan)#name ADMIN
SW-ADMIN(config-vlan)#exit
SW-ADMIN(config)#
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
SW-ADMIN(config)#vlan 99
SW-ADMIN(config-vlan)#name NETWORK_MANAGEMENT
SW-ADMIN(config-vlan)#exit
SW-ADMIN(config)#
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
SW-ADMIN(config)#interface range fastEthernet0/1 - 3
SW-ADMIN(config-if-range)#switchport mode access
% Invalid input detected at '' marker.
SW-ADMIN(config-if-range)#
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/2 (99).
SW-ADMIN(config-if-range)#switchport mode access
SW-ADMIN(config-if-range)#switchport access vlan 10
SW-ADMIN(config-if-range)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
SW-ADMIN(config-if-range)#no shutdown
SW-ADMIN(config-if-range)#exit
SW-ADMIN(config)#
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/2 (1), with CORE-SW GigabitEthernet1/0/2 (99).
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/2 (1), with CORE-SW GigabitEthernet1/0/2 (99).
Copy Paste
Top
```

B2. SW-ADMIN VLAN 10, VLAN 99 and access-port configuration.

```
SW-SALES
Physical Config CLI Attributes
IOS Command Line Interface

SW-SALES#en
SW-SALES#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW-SALES(config)#vlan 20
SW-SALES(config-vlan)#name SALES_POS
SW-SALES(config-vlan)#exit
SW-SALES(config)#
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/3 (99).
SW-SALES(config)#vlan 99
SW-SALES(config-vlan)#name NETWORK_MANAGEMENT
SW-SALES(config-vlan)#exit
SW-SALES(config)#interface range fastEthernet0/1 - 4
SW-SALES(config-if-range)#switchport mode access
SW-SALES(config-if-range)#
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/3 (99).
SW-SALES(config-if-range)#switchport access vlan 20
SW-SALES(config-if-range)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
SW-SALES(config-if-range)#no shutdown
SW-SALES(config-if-range)#exit
SW-SALES(config)#
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/3 (99).
SW-SALES(config)#interface gigabitEthernet0/1
SW-SALES(config-if)#description TO-CORE-SW
SW-SALES(config-if)#switchport mode trunk
SW-SALES(config-if)#switchport trunk native vlan 99
SW-SALES(config-if)#SPANNTREE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/1 on VLAN0099. Port consistency restored.
%SPANNTREE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/1 on VLAN0001. Port consistency restored.
Copy Paste
Top
```

B3. SW-SALES VLAN 20 and trunk/native VLAN configuration.

```
SW-WAREHOUSE
Physical Config CLI Attributes
IOS Command Line Interface

SW-WAREHOUSE>en
SW-WAREHOUSE#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW-WAREHOUSE(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/4 (99).
SW-WAREHOUSE(config)#vlan 30
SW-WAREHOUSE(config-vlan)#name WAREHOUSE
SW-WAREHOUSE(config-vlan)#exit
SW-WAREHOUSE(config)#vlan 99
SW-WAREHOUSE(config-vlan)#name NETWORK_MANAGEMENT
SW-WAREHOUSE(config-vlan)#exit
SW-WAREHOUSE(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/4 (99).
SW-WAREHOUSE(config)#interface range fastEthernet0/1 - 2
SW-WAREHOUSE(config-if-range)#switchport mode access
SW-WAREHOUSE(config-if-range)#switchport access vlan 30
SW-WAREHOUSE(config-if-range)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
SW-WAREHOUSE(config-if-range)#no shutdown
SW-WAREHOUSE(config-if-range)#exit
SW-WAREHOUSE(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with CORE-SW GigabitEthernet1/0/4 (99).
SW-WAREHOUSE(config)#interface gigabitEthernet0/1
SW-WAREHOUSE(config-if)#description TO-CORE-SW
SW-WAREHOUSE(config-if)#switchport mode trunk
SW-WAREHOUSE(config-if)#switchport trunk native vlan 99
SW-WAREHOUSE(config-if)#%SFANTRIE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/1 on VLAN999. Port consistency restored.
%SFANTRIE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/1 on VLAN9001. Port consistency restored.
SW-WAREHOUSE(config-if)#switchport trunk allowed vlan 30,99
SW-WAREHOUSE(config-if)#no shutdown
SW-WAREHOUSE(config-if)#exit
SW-WAREHOUSE(config)#interface vlan 99
SW-WAREHOUSE(config-vlan)#
VLAN99-5-CHANGED: Interface Vlan99, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
SW-WAREHOUSE(config-if)#ip address 172.30.2.164 255.255.255.240
SW-WAREHOUSE(config-if)#no shutdown
SW-WAREHOUSE(config-if)#exit
SW-WAREHOUSE(config)#ip default-gateway 172.30.2.161
```

B4. SW-WAREHOUSE VLAN 30, VLAN 99 and trunk configuration.